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Question Based on Array in C

1. Write a program to input and print all elements of an array.
2. Write a program to find the sum of all elements in an array.
3. Write a program to find sum of array elements & check sum is even or odd.
4. Write a program to find the average of array elements.
5. Write a program to display all even numbers present in an array.
6. Write a program to display all odd numbers present in an array.
7. Write a program to display all positive numbers present in an array.
8. Write a program to display all negative numbers present in an array.
9. Write a program to display all elements in an array that are divisible by 4.
10. Write a program to display all elements in an array that are multiples of 3.
11. Write a program to count how many even numbers are present in an array.
12. Write a program to count how many odd numbers are present in an array.
13. Write a program to count how many positive numbers are present in an array.
14. Write a program to count how many negative numbers are present in an array.

15. Write a program to count how many elements in an array are divisible by 4.
16. Write a program to count how many elements in an array are multiples of 3.
17. Write a program to print squares of all numbers present in a given array.
18. Write a program to print cube of all numbers present in a given array.
19. Write a program to display the array elements in reverse order.
20. Write a program to copy elements from one array to another.
21. Write a program to find the maximum element in an array.
22. Write a program to find the minimum element in an array.
23. Write a program that accepts some integers from the user and finds the highest value and the input position.
24. Write a program to input elements into an array and divide them into two separate arrays — one containing even numbers and the other containing odd numbers.
25. Write a program to input elements into an array and divide them into two separate arrays — one containing positive numbers and the other containing negative numbers.
26. Write a program to take array elements from the user, store their squares in a new array, and display the new array.

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27. Write a program to take array elements from the user, add 5 to each element, store the results in a new array, and display the new array.
28. Write a program to count how many elements in an array are prime and how many are not prime.
29. Write a program to count how many elements in an array are armstrong and how many are not armstrong.
30. Write a program to count how many elements in an array are perfect and how many are not perfect.
31. Write a program to count how many elements in an array are prime and how many are not prime.
32. Write a program to count how many elements in an array are divisible by 3, 5, and 8.
33. Write a program to input elements into an array and divide them into two separate arrays — one containing prime numbers and the other containing not prime numbers.
34. Write a program to interchange first and last elements in an array.
35. Write a program to check (search) given number is present or not present in array, using Linear search.
36. Write a program to check (search) given number is present or not present in array, using binary search.
37. Write a program to reverse all elements of an array without using any extra array.
38. Write a program to check how many digits at each index of array.

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39. Write a program to reverse the digits of every element present in an array.
40. Write a program to arrange (sort) array elements in ascending or descending order.
41. Write a program to check given array is sorted or not.
42. Write a program to arrange (sort) array elements using Selection Sort.
43. Write a program to arrange (sort) array elements using Bubble Sort.
44. Write a program to find the second largest and second smallest element.
45. Write a program to merge any two array in a single array.
46. Write a program to merge any two sorted array in a single sorted array .
47. Write a program to find how many times a given number appears in an array.
48. Write a program to calculate and print the sum of squares of all prime numbers present in a given array.
49. Write a program to take array elements from the user, replace each element with the sum of its individual digits, and display the updated array.
50. Write a program to change array element which element have digit 2 by 3.
51. Write a program to take array elements from the user, shift each element one position to the left, store the result in a new array, and display the new array.
(Hint: The first element becomes the last element, i.e., $A1[i] = a[(i + 1) \% n]$.)

52. Write a program to take array elements from the user, replace every occurrence of 5 with 0, shift all zeros to the end of the array, and create a new array with the updated values.
53. Write a program to create a new array from a given array of integers shifting all zeros to left direction
54. Write a program to create an array from a given array of integers shifting all even numbers before all odd numbers
55. Write a program to read an array of length 6, change the first element by the last, the second element by the fifth and the third element by the fourth. Print the elements of the modified array
56. Write a program to take array elements from the user and count the total number of elements that appear more than once in the array (duplicate elements count).
57. Write a program to take array elements from the user and count how many elements appear exactly once in the array.
58. Write a program to print all unique elements in an array.
59. Write a program to take array elements from the user, remove all duplicate values, and display the array containing only distinct elements.

60. Write a program to take array elements from the user and display how many times each element appears in the array(count the frequency of each element).
61. Write a program to take a sorted array and a value from the user, find the correct position for the value to maintain the sorted order, insert it at that position, and display the updated array.
62. Write a program to take array elements from the user, insert a given element at a specified position, and display the updated array.
63. Write a program to take array elements from the user, delete a specified element, and display the updated array.
64. Write a program to take array elements from the user, delete the element at a given position, and display the updated array.
65. Write a program to take array elements from the user and find the maximum product that can be obtained by multiplying any two elements in the array.
66. Write a program to take array elements from the user, calculate the average of the elements, and count how many elements are greater than this average.

67. Write a program to take array elements from the user and rotate the array either to the left or right by a given number of positions k , then display the rotated array.

68. Write a program to take two arrays of the same size from the user and create a new array where each element is the sum of the corresponding elements from the two arrays.

69. Write a program to take two arrays of the same size from the user and determine whether they are identical (all corresponding elements are equal) or different.

70. Write a program to take an unsorted array from the user, allow the user to specify a value and a position, insert the value at that position, and display the updated array.

71. Write a program to take array elements from the user and determine whether the array reads the same forwards and backwards (i.e., it is a palindrome) or not.

72. Write a program to take array elements and a target sum from the user, and find if there exists a pair of elements in the array whose sum is equal to the given target.

73. Write a program to take array elements from the user and find the majority element, if it exists.

Note: A majority element in an array of size n is an element that appears more than $n/2$ times. There can be at most one such element.

74. Write a program to take array elements from the user where all numbers occur an even number of times except one number that occurs an odd number of times, and find that number.

75. Write a program to take array elements from the user and find the maximum sum that can be obtained from any contiguous subarray.

(Hint: This is also known as the Maximum Subarray Problem, which can be solved using Kadane's Algorithm.)

76. Write a program to take array elements from the user where all numbers are distinct and in a known range (for example, 1 to n), and find the number that is missing from the array.

77. Write a program to find the ceiling in a sorted array.

78. Write a program to find the floor in a sorted array.

79. Write a program to take array elements from the user and find the pair of elements whose sum is closest to zero.

80. Write a program to take elements of a list from the user and determine which element appears the most number of times (the mode of the list).

81.If an element has less than 4 digits, append 9s at the end to make it 4 digits.

Example:

Input list: [123, 5432, 6, 89, 1432]

Output list: [1239, 5432, 6999, 8999, 1432]

82.Each sublist contains the individual digits of the corresponding number from the original list, excluding all even digits.

Example:

Input list: [1352, 596, 467, 8932]

Output list: [[1, 3, 5], [5, 9], [7], [9, 3]]

83.Write a Python program to count the number of strings from a given list of strings. The string length is 2 or more and the first and last characters are the same.

Sample List : ['abc', 'xyz', 'aba', '1221']

Expected Result : 2

84.Write a Python program to take a list of non-empty tuples and return a new list sorted in increasing order based on the last element of each tuple.

Example:

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Input: [(2, 5), (1, 2), (4, 4), (2, 3), (2, 1)]

Output: [(2, 1), (1, 2), (2, 3), (4, 4), (2, 5)]

85. Write a Python program to check whether a list is empty and print "empty" if it is.

86. Write a Python function that takes two lists and returns True if they have at least one common member.

87. Write a Python program to take a dictionary where each key maps to a list of values, and create a new dictionary where each value maps to a list of keys that contain it.

Example:

Input: dict1 = {"a": [1, 2], "b": [1], "c": [2]}

Output: dict2 = {1: ["a", "b"], 2: ["a", "c"]}

88. Write a Python program to take a list of lists and create a new list where each element is the sum of the corresponding sublist.

Example:

Input: l1 = [[1, 2, 3], [1, 2, 3], [1, 2, 3]]

Output: l2 = [6, 6, 6]

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